

Multiple Choice (10 marks)

1. The difference between dc and ac in electric circuits is that in dc, charges flow
 - (a) to and fro
 - (b) in the direction opposite to the flow of current
 - (c) in a circular direction
 - (d) only when the circuit is closed
 - (e) in a random direction
2. A camera objective of 50 mm focal length is set at aperture stop ($f / 4$). If the camera is then focused at an object 20 cm away what will the effective f-number be?
 - (a) 7.1
 - (b) 9.4
 - (c) 4.0
 - (d) 3.8
 - (e) 5.33
3. What is the second most common element in the solar system?
 - (a) Methane
 - (b) Nitrogen
 - (c) Oxygen
 - (d) Iron
 - (e) Hydrogen
4. Colors in a soap bubble result from light
 - (a) dispersion
 - (b) deflection
 - (c) refraction
 - (d) reflection
 - (e) interference
5. An electric iron with a resistance of 24 ohms is connected to a 120-volt line. What power is being used?
 - (a) 600 watts
 - (b) 800 watts

- (c) 750 watts
- (d) 1000 watts
- (e) 700 watts

True / False (10 marks)

Answer True or False

6. True or False: When pulling a rope down with 25 N of force in an ideal pulley system, the crate rises 25 cm for every 1 m of rope pulled.
7. True or False: A magnetic force can change the charge of a beam of electrons as it interacts with a magnetic field.
8. True or False: The minimum thickness of crystalline quartz required for a quarter wave plate at 589 nm is 0.016 mm.
9. True or False: A freight engine with 400 hp can pull a string of freight cars weighing 200 tons at a maximum speed of 75 mi/hr under the given conditions.
10. True or False: When the temperature of a strip of iron is increased, it may decrease in length under certain conditions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A 100-kg man jumps into a swimming pool from a height of 5 m. Analyze the forces at play and calculate the average force exerted by the water during the 0.4 seconds it takes to stop him.
12. Consider a meter stick traveling at a speed of $0.8c$ relative to an observer. Discuss how to calculate the time it takes for the stick to pass the observer, incorporating concepts of relativistic length contraction and time dilation.

End of Paper